Liquids – Clearly Unique!: Investigating the Physical Properties of Water, Mineral Oil, and Corn Syrup

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Abstract

This experiment investigated the physical properties of three common clear, colorless liquids — water, mineral oil, and corn syrup — through two simple observational tests. In the first test, small drops of each liquid were placed on a plastic surface inside a zip-closing bag and the bag was tilted to observe how each liquid moved. In the second test, one drop of food coloring was added to each liquid to observe how the coloring dispersed. The results revealed distinct differences in viscosity, surface tension, and molecular interactions among the three liquids. Water formed the most spherical droplet and ran quickly down the plastic surface; mineral oil spread flat and moved slowly, leaving a streak; and corn syrup barely moved at all. When food coloring was added, it sank and spread in water, sank but clumped in oil, and remained on top and spread in corn syrup. These observations are explained by the different molecular structures and intermolecular forces present in each liquid.

1. Introduction

Many liquids share the appearance of being clear and colorless, yet they can behave very differently due to their distinct molecular compositions and intermolecular forces. Water (H■O) is one of the most studied liquids due to its unique properties, including high surface tension, polarity, and hydrogen bonding. Mineral oil, a non-polar hydrocarbon mixture, and corn syrup, a concentrated sugar solution, also appear visually similar to water but have very different physical properties.

This experiment explores and compares the physical properties of water, mineral oil, and corn syrup using two straightforward tests: a droplet-on-plastic tilt test and a food-coloring dispersion test. These tests provide qualitative evidence of differences in viscosity, surface tension, and polarity among the three liquids.

2. Theory

The behavior of a liquid on a surface and its interaction with other substances is governed by its intermolecular forces and molecular polarity. Three key concepts apply to this experiment:

Surface Tension: Surface tension arises from cohesive forces between liquid molecules at the surface. Water has a particularly high surface tension due to hydrogen bonding between its polar molecules (see Figure 1). This causes water droplets to bead up into a more spherical shape on non-polar surfaces such

as plastic.

Viscosity: Viscosity is the resistance of a liquid to flow, resulting from internal friction between its molecules. Corn syrup has a very high viscosity due to its large, entangled sugar molecules (primarily glucose and fructose polymers), causing it to flow very slowly. Mineral oil has intermediate viscosity, while water is comparatively much less viscous.

Polarity and Miscibility: Water is a polar solvent that readily dissolves other polar substances (like food coloring dyes) through dipole-dipole interactions and hydrogen bonding. Mineral oil is non-polar and does not interact favorably with polar molecules. Corn syrup is a polar, aqueous-based solution that is highly viscous, allowing food coloring to spread across its surface slowly rather than sinking.

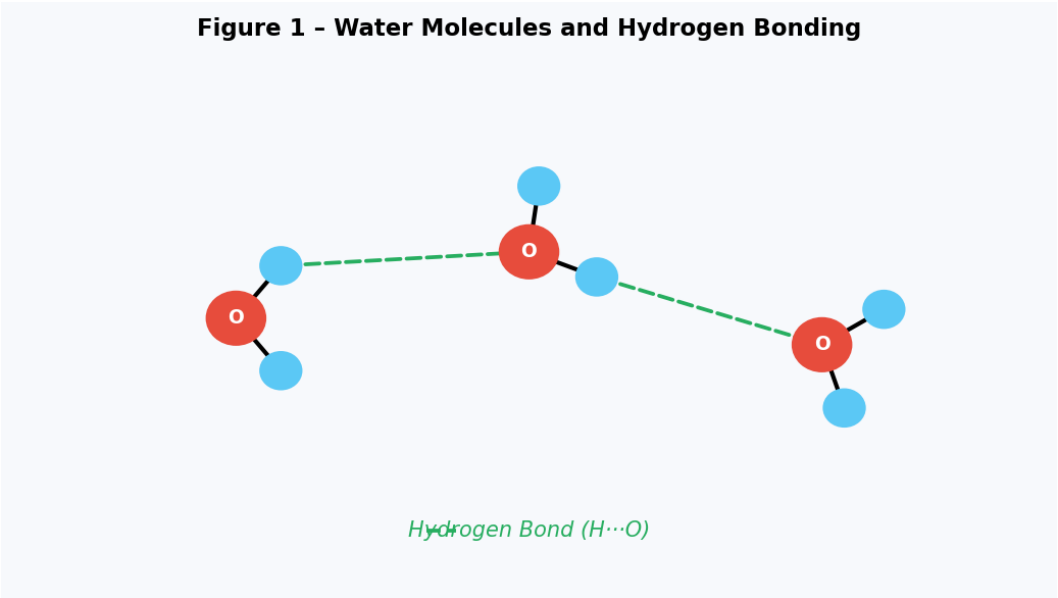

Figure 1 – Water molecules connected by hydrogen bonds (dashed green lines). The partial positive charges (δ+) on hydrogen and partial negative charge (δ–) on oxygen create strong intermolecular attractions.

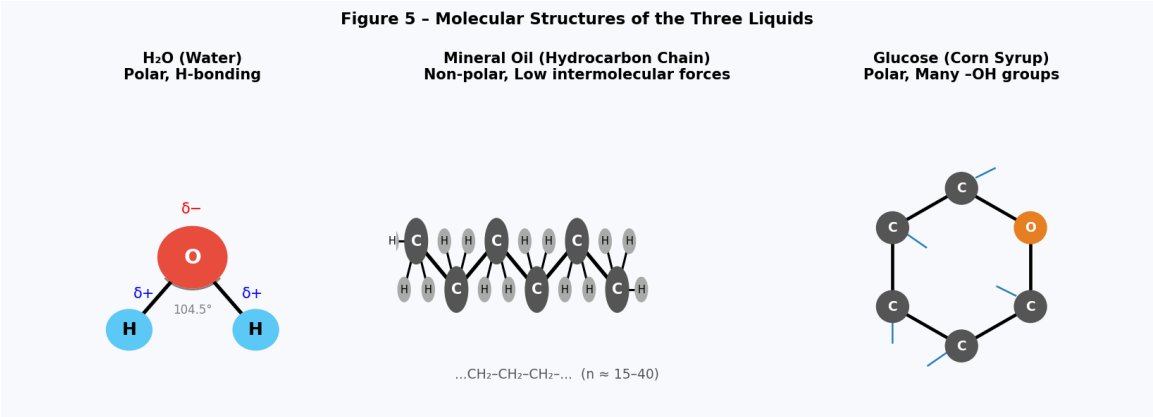

Figure 2 – Molecular structure comparison: water (polar, H-bonding), a hydrocarbon chain typical of mineral oil (non-polar), and the glucose ring found in corn syrup (polar, many –OH groups).

3. Materials and Methods

Materials:

• 3 clear plastic cups

• Masking tape and pen (for labeling)

• Water

• Mineral oil

• Corn syrup

• Straws or droppers

• Zip-closing plastic bag

• Stiff card paper (cardboard)

• Food coloring (one color)

• Safety goggles

Safety:

Food coloring may stain skin and clothing. All work was conducted with adult supervision. Both the experimenter and the supervising adult wore properly fitting safety goggles throughout the experiment. No liquids were ingested.

Procedure – Test 1 (Tilt Test):

1. Label three cups: Water, Oil, and Corn Syrup using masking tape and a pen.

2. Place approximately ½ teaspoon of each liquid into its labeled cup.

3. Label three areas on a stiff card inside a zip-closing plastic bag and seal the bag.

4. Using separate droppers or straws, place one drop of each liquid onto the plastic in its labeled area.

5. Observe the shape of each droplet.

6. Tilt the plastic bag and observe how each liquid moves down the surface.

Procedure – Test 2 (Food Coloring Dispersion Test):

1. Add more of each liquid to their labeled cups until approximately half full.

2. Hold the food coloring container close to the surface of the water and carefully add one drop.

3. Repeat step 2 for the oil and the corn syrup.

4. Observe and record how the food coloring behaves in each liquid.

4. Observations

Test 1: Droplet Shape and Tilt Behavior

When the drops were first placed on the plastic surface, noticeable differences in droplet shape were observed. The water droplet was the most spherical and bead-like among the three, indicating high surface tension and poor adhesion to the non-polar plastic. The mineral oil droplet was considerably flatter and spread out more across the surface. The corn syrup droplet was intermediate in shape — somewhat spherical like water but slightly flatter (Figure 3).

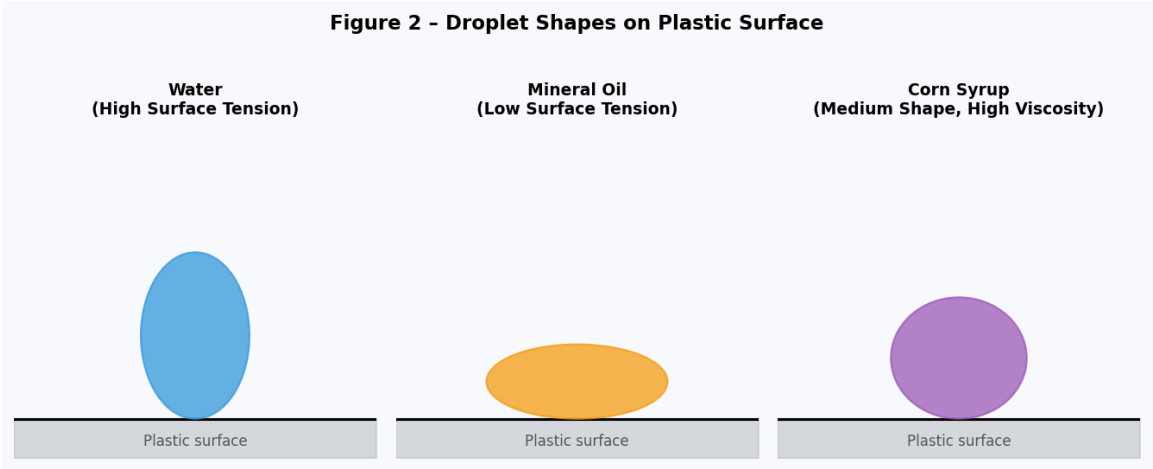

Figure 3 – Droplet shapes on plastic: water beads up (high surface tension), mineral oil flattens (low surface tension), corn syrup is intermediate.

When the plastic bag was tilted, differences in viscosity became apparent. The water ran down the surface very quickly and left virtually no visible residue. The mineral oil moved slowly and left a long, greasy streak on the plastic. The corn syrup barely moved at all (Figure 4).

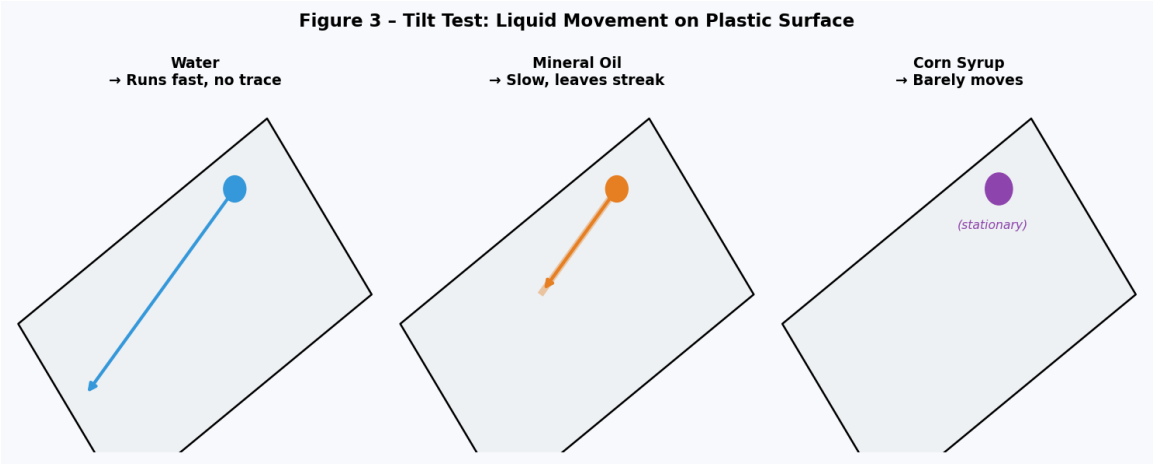

Figure 4 – Tilt test results: water runs rapidly with no trace, mineral oil moves slowly leaving a streak, and corn syrup remains nearly stationary.

Liquid	Droplet Shape	Movement on Tilt	Residue Left
Water	Spherical (ball-like)	Very fast, no trace	None
Mineral Oil	Flat, spread out	Slow	Long oily streak
Corn Syrup	Slightly flattened sphere	Barely moved	Remained in place

Table 1 – Droplet shape and tilt behavior for each liquid.

Test 2: Food Coloring Dispersion

When one drop of food coloring was added to each liquid, distinct behaviors were observed. In the water, the food coloring sank slowly and then spread outward, gradually diffusing throughout the liquid. In the mineral oil, the food coloring sank directly to the bottom but stayed together in a compact droplet rather than dispersing. In the corn syrup, the food coloring remained on the top surface and spread outward very slowly (Figure 5).

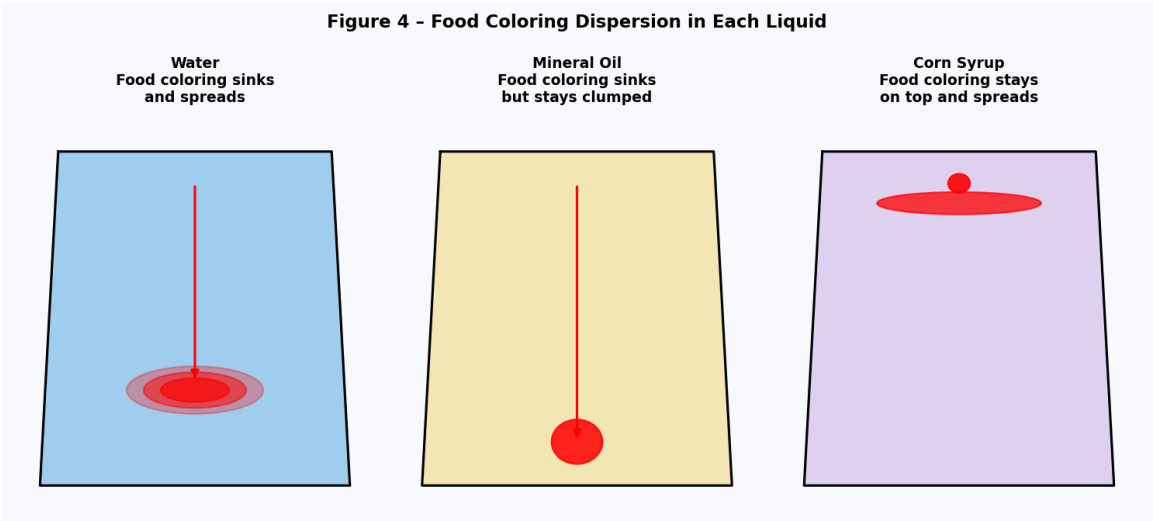

Figure 5 – Food coloring dispersion: disperses in water, clumps at the bottom in mineral oil, and spreads on the surface of corn syrup.

Liquid	Food Coloring Behavior
Water	Sank and spread outward; dispersed throughout
Mineral Oil	Sank to the bottom; stayed together as a clump
Corn Syrup	Remained on top; spread out slowly on the surface

Table 2 – Food coloring behavior in each liquid.

5. Discussion

The observed differences between the three liquids can be explained by their molecular properties. Water is a small, polar molecule that forms extensive hydrogen bonds (Figure 1), giving it high surface tension and low viscosity. Its polarity also makes it an excellent solvent for food coloring, which consists of polar dye molecules, explaining why the food coloring dispersed readily in water.

Mineral oil is composed of long, non-polar hydrocarbon chains (Figure 2). Its non-polar nature accounts for its poor interaction with the polar plastic surface and with food coloring, causing the coloring to remain as a tight droplet rather than dispersing. Its moderate viscosity explains why it moved more slowly down the plastic surface and left a streak.

Corn syrup is an aqueous solution of sugars (primarily glucose polymers) and has a very high viscosity due to the large, tangled polymer chains that resist flow. Its polar, aqueous base makes it compatible with food coloring, but its high viscosity prevents rapid mixing, so the food coloring spread only on the surface rather than sinking and dispersing. Its density, which is greater than that of the food coloring drop, also contributes to keeping the coloring at the surface.

These results demonstrate how the molecular composition of a liquid directly governs its macroscopic physical properties such as surface tension, viscosity, and miscibility. Even liquids that appear identical

to the naked eye can behave in fundamentally different ways based on their molecular structure.

6. Conclusion

This experiment successfully demonstrated that water, mineral oil, and corn syrup, despite all appearing as clear and colorless liquids, have significantly different physical properties. The tilt test revealed differences in surface tension and viscosity: water moved fastest with no residue, oil moved slowly and left a streak, and corn syrup barely moved. The food coloring test highlighted differences in polarity and molecular interactions: food coloring dispersed in water, clumped in oil, and spread on top of corn syrup.

These findings underscore the importance of molecular properties in determining the behavior of liquids in real-world situations. The experiment provided clear, observable evidence that macroscopic properties of liquids — such as how fast they flow or how they interact with other substances — are directly tied to the nature of their constituent molecules and the intermolecular forces at play.

References

American Chemical Society. (2019). *Adventures in Chemistry: Liquids – Clearly Unique!* Retrieved from www.acs.org/kids